

Problem Set 6 – Mathematical induction

Session 6 · Ordinary induction, strong induction, sums, divisibility, and common errors

HOW TO ANSWER

Every answer carries one sentence of justification. A value on its own is not a solution – say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

NOTATION

$d \mid N$ means that $N = dq$ for some integer q . In words, " d divides N " means that N is a multiple of d .

FOR EACH INDUCTION PROOF, MAKE THESE PARTS EASY TO FIND

1. the statement $P(n)$ and its starting value;
2. the base case;
3. the inductive hypothesis for an arbitrary allowed integer k ;
4. the inductive step from $P(k)$ to $P(k + 1)$;
5. the conclusion by induction.

Part A · Start with the structure

1 NAME THE THREE MAIN PARTS

For the claim

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad (n \geq 1),$$

write:

1. the base-case statement $P(1)$;
2. the inductive hypothesis $P(k)$;
3. the exact statement $P(k+1)$ that you must prove.

Do not prove the claim in this question. Write the three statements clearly.

SOLUTION

$$P(1) : 1 = \frac{1(1+1)}{2}.$$

$$P(k) : 1 + 2 + \cdots + k = \frac{k(k+1)}{2} \quad \text{for some } k \geq 1.$$

$$P(k+1) : 1 + 2 + \cdots + k + (k+1) = \frac{(k+1)(k+2)}{2}.$$

2 SUM OF ODD NUMBERS

Prove by induction that, for every integer $n \geq 1$,

$$1 + 3 + 5 + \cdots + (2n-1) = n^2.$$

In your inductive step, explain why the next term is $2k+1$.

SOLUTION Let $P(n)$ be

$$1 + 3 + 5 + \cdots + (2n-1) = n^2.$$

Base case: for $n = 1$, the left side is 1 and the right side is $1^2 = 1$.

Inductive hypothesis: let $k \geq 1$ and assume

$$1 + 3 + \cdots + (2k-1) = k^2.$$

The next term is $2(k+1) - 1 = 2k+1$. Therefore

$$\begin{aligned} 1 + 3 + \cdots + (2k-1) + (2k+1) &= k^2 + 2k + 1 \\ &= (k+1)^2. \end{aligned}$$

This proves $P(k+1)$. By induction, $P(n)$ is true for every integer $n \geq 1$. $\square$

3 AN EXPONENTIAL INEQUALITY

Prove by induction that, for every integer $n \geq 0$,

$$2^n \geq n + 1.$$

In the inductive step, state why $2(k + 1) \geq k + 2$.

SOLUTION Let $P(n)$ be $2^n \geq n + 1$ for $n \geq 0$.

Base case: $2^0 = 1 \geq 0 + 1$.

Inductive hypothesis: let $k \geq 0$ and assume $2^k \geq k + 1$. Then

$$2^{k+1} = 2 \cdot 2^k \geq 2(k + 1).$$

Because $k \geq 0$, we have $2(k + 1) = 2k + 2 \geq k + 2$. Thus $2^{k+1} \geq k + 2$, which is $P(k + 1)$. By induction, $2^n \geq n + 1$ for every integer $n \geq 0$. $\square$

Part B · Write complete proofs

4 DIVISIBILITY WITH AN EXPONENTIAL

Prove by induction that, for every integer $n \geq 1$,

$$5 \mid (6^n - 1).$$

Write the inductive hypothesis in the form $6^k - 1 = 5m$ for some integer m . In the next case, end with 5 times an integer.

SOLUTION Let $P(n)$ be $5 \mid (6^n - 1)$ for $n \geq 1$.

Base case: $6^1 - 1 = 5 = 5 \cdot 1$, so $P(1)$ is true.

Inductive hypothesis: let $k \geq 1$ and assume $6^k - 1 = 5m$ for some integer m . Then

$$\begin{aligned} 6^{k+1} - 1 &= 6 \cdot 6^k - 1 \\ &= 6(6^k - 1) + 5 \\ &= 6(5m) + 5 \\ &= 5(6m + 1). \end{aligned}$$

Since $6m + 1$ is an integer, $5 \mid (6^{k+1} - 1)$. By induction, $5 \mid (6^n - 1)$ for every integer $n \geq 1$. $\square$

5 SUM OF SQUARES

Prove by induction that, for every integer $n \geq 1$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

In the inductive step, factor out $k+1$ before simplifying.

SOLUTION Let $P(n)$ be

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for $n \geq 1$.

Base case:

$$1^2 = 1 = \frac{1 \cdot 2 \cdot 3}{6}.$$

Inductive hypothesis: let $k \geq 1$ and assume

$$1^2 + 2^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}.$$

Then

$$\begin{aligned} 1^2 + \cdots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{(k+1)(k(2k+1) + 6(k+1))}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6}. \end{aligned}$$

This is the formula with $n = k+1$. By induction, $P(n)$ is true for every integer $n \geq 1$. $\square$

Prove by induction that, for every integer $n \geq 1$,

$$3 \mid (n^3 - n).$$

Write the inductive hypothesis in the form $k^3 - k = 3m$ for some integer m .

SOLUTION Let $P(n)$ be $3 \mid (n^3 - n)$ for $n \geq 1$.

Base case: $1^3 - 1 = 0 = 3 \cdot 0$.

Inductive hypothesis: let $k \geq 1$ and assume $k^3 - k = 3m$ for some integer m . Then

$$\begin{aligned}(k+1)^3 - (k+1) &= (k^3 - k) + 3k(k+1) \\ &= 3m + 3k(k+1) \\ &= 3(m + k(k+1)).\end{aligned}$$

The number in brackets is an integer. Thus $3 \mid ((k+1)^3 - (k+1))$. By induction, $3 \mid (n^3 - n)$ for every integer $n \geq 1$. $\square$

Prove by **strong induction** that every integer $n \geq 12$ can be written as

$$n = 4a + 5b$$

for non-negative integers a and b . Here, non-negative means $0, 1, 2, \dots$

1. Show the four base cases $n = 12, 13, 14, 15$.
2. For the step, let $k \geq 15$ and assume that every integer from 12 to k has this form.
3. Use the earlier number $k - 3$ to prove the result for $k + 1$.

State why the strong hypothesis includes $k - 3$.

SOLUTION Let $P(n)$ mean that $n = 4a + 5b$ for some non-negative integers a, b .

Base cases:

$$12 = 3 \cdot 4, \quad 13 = 2 \cdot 4 + 5, \quad 14 = 4 + 2 \cdot 5, \quad 15 = 3 \cdot 5.$$

Strong inductive hypothesis: let $k \geq 15$ and assume that $P(m)$ is true for every integer m with $12 \leq m \leq k$. Since $k \geq 15$,

$$12 \leq k - 3 \leq k.$$

So the hypothesis applies to $k - 3$. Write $k - 3 = 4a + 5b$ for some non-negative integers a, b . Then

$$k + 1 = (k - 3) + 4 = 4(a + 1) + 5b.$$

The integers $a + 1$ and b are non-negative. Thus $P(k + 1)$ is true. By strong induction, every integer $n \geq 12$ has the required form. $\square$

Part C · Read a proof critically

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THE "ALL HORSES HAVE THE SAME COLOUR" ERROR

Someone tries to prove the false claim below by induction.

THE CLAIM Claim $P(n)$: Every group of n horses has horses of only one colour.

Base case: A group of one horse has only one colour.

Inductive step: Assume $P(k)$. Take a group of $k + 1$ horses. The first k horses have one colour by $P(k)$. The last k horses also have one colour by $P(k)$. These two groups share $k - 1$ horses, so all $k + 1$ horses have the same colour.

Answer all three questions.

1. Which value of k makes the inductive step fail?
2. Why does the "shared horses" sentence fail at this value?
3. Explain why adding more base cases cannot prove the original claim.

SOLUTION

1. The step fails at $k = 1$.
2. At $k = 1$, the first group of k horses and the last group of k horses share $k - 1 = 0$ horses. There is no shared horse that links the colours of the two one-horse groups.
3. More base cases cannot prove the original claim, because the claim itself is false: a group of two horses can contain horses of different colours. The step from $k = 1$ to $k = 2$ must be valid, and it is not.

SELF-CHECK BEFORE SUBMITTING

1. Did I write the correct starting value for every proof?
2. Did I state an inductive hypothesis for one allowed integer k ?
3. Did I use that hypothesis in the step?
4. Did I prove the exact next statement, $P(k + 1)$?
5. Did I write a conclusion that gives the correct range of n ?

